Applied Mathematics 1

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How to use these notes

These notes are rewritten from the supplied MATH0008 typeset handouts and handwritten lecture pages dated 5 October–6 November 2026. They are a student study companion, not an official or verbatim set of lecture notes. The original sources remain in `lecture_resources/`.

The chapter order follows the handouts: discrete dynamical systems, continuous models, then mechanics. Each model is presented through its assumptions, equations, method of analysis and interpretation. Algebraic steps omitted in the sources are filled in, and important qualifications are stated explicitly. Additional explanations are labelled where they go beyond the supplied derivation.

Source examples are identified where useful, but numbering in this edition is automatic and need not match the originals.

Some typed material is ahead of the supplied handwritten pages: notably parts of SIR analysis and the early mechanics examples. It is included as handout material, not claimed to have been covered in a particular lecture. Any optional or non-examinable label refers only to the supplied source; check the current course for assessment scope.

Figures are newly drawn explanatory diagrams, not copies of source scans. They are kept separately in `figures/` to keep the chapter source readable. The README records the source inventory and principal corrections.

Notation. In a discrete model, $t = 0, 1, 2, \ldots$ counts generations. In a continuous model, $\dot{x} = dx/dt$ and $\ddot{x} = d^2x/dt^2$. An equilibrium is denoted by a superscript $*$; a subscript 0 denotes an initial value. Populations and compartment sizes must be nonnegative. Unless stated otherwise, model parameters described as rates are positive.

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Chapter 1

Discrete Dynamical Systems

1.1 Building a discrete population model

A discrete-time model describes a state at successive time steps:

$$x_0,\ x_1,\ x_2, \ldots, \qquad x_{t+1} = f(x_t).$$

It is appropriate, for example, for a population with distinct annual breeding seasons. The initial condition x_0 must be specified. Population values are treated as real expected sizes or large-population approximations; the recurrence need not give integer outputs. A physically meaningful population must satisfy $N_t \geq 0$.

1.1.1 Constant reproduction rate

Suppose every individual produces on average $r \geq 0$ offspring and the old generation is replaced completely. Then

$$N_{t+1} = rN_t, \qquad N_t = r^t N_0.$$

Repeated substitution or induction proves the solution. If $N_0 > 0$:

- $0 \leq r < 1$: the population tends to zero;
- $r = 1$: the population stays constant;
- $r > 1$: it grows without bound.

For $r > 0$, $r^t = e^{t \log r}$, hence the description *exponential* growth or decay. Time is still discrete even when the formula uses an exponential function.

1.1.2 Fixed harvesting

If $h \geq 0$ individuals are removed after reproduction, the algebraic model is

$$N_{t+1} = rN_t - h.$$

For $r \neq 1$, iteration gives

$$\begin{aligned} N_t &= r^t N_0 - h(1 + r + \cdots + r^{t-1}) \\ &= r^t N_0 - h\frac{r^t - 1}{r - 1} \\ &= N^* + r^t(N_0 - N^*), \qquad N^* = \frac{h}{r-1}. \end{aligned}$$

For $r = 1$, $N_t = N_0 - th$.

If $r > 1$ and $h > 0$, $N^* > 0$ is a threshold:

- $N_0 > N^*$ produces unbounded growth in this model;
- $N_0 = N^*$ produces a constant population;
- $N_0 < N^*$ eventually makes the algebraic formula nonpositive.

Negative population predictions indicate failure of the unmodified model. The 5 October handwritten notes explicitly repair it as

$$N_{t+1} = \max\{rN_t - h, 0\}.$$

Under this repaired model, a below-threshold population reaches zero in finite time and remains extinct. The equality case $N_0 = N^*$ is an equilibrium, not extinction. For $h = 0$, this reduces to reproduction alone.

Example 1.1 (Checking a harvesting threshold)**.** As an added illustration, let $r = 2$ and $h = 10$. Then $N^* = 10$. Starting at 12 gives $N_t = 10 + 2^{t+1}$; starting at 10 keeps it at 10. Starting at 8 gives $8, 6, 2, 0, 0, \ldots$ in the clipped model. The unclipped next value after 2 would be -6, which is not a population.

1.2 The logistic map and nondimensionalisation

To represent reduced reproduction at high density, take

$$N_{t+1} = rN_t\left(1 - \frac{N_t}{K}\right), \qquad K > 0.$$

Set $x_t = N_t/K$. Dividing by K gives the *logistic map*

$$x_{t+1} = rx_t(1 - x_t) = f(x_t). \tag{1.1}$$

Both x_t and r are dimensionless. Scaling removes K from the recurrence, leaving one parameter controlling the dynamics.

Remark (What K means here)*.* In this discrete model, K is the density scale where the next-generation reproduction factor vanishes. The nonzero equilibrium is $N^* = K(1 - 1/r)$, not K. This differs from the continuous logistic model, whose carrying capacity is itself the positive equilibrium.

For $0 \le x \le 1$, $0 \le x(1 - x) \le 1/4$. Consequently, if $0 \le r \le 4$, the interval $[0, 1]$ is invariant: an initial value there stays there. For $0 < r < 4$ and $0 < x_0 < 1$, subsequent values remain in $(0, 1)$. For larger r, some initial values leave the biologically meaningful range.

1.3 Fixed points and cobweb diagrams

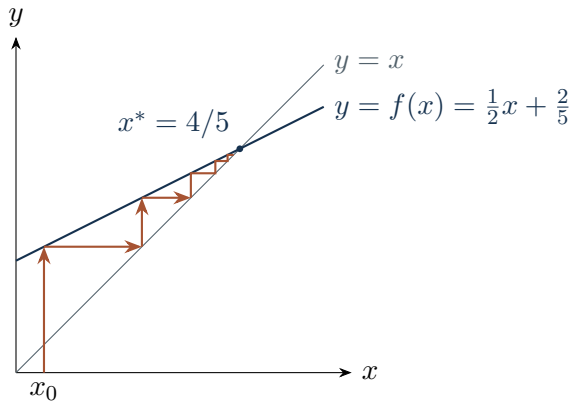
Definition 1.1 (Fixed point)**.** A fixed point of $x_{t+1} = f(x_t)$ is a value x^* with $f(x^*) = x^*$. Starting exactly at x^* gives a constant sequence. It is also called a steady state or an equilibrium of the map.

Fixed points are intersections of $y = f(x)$ with $y = x$. For the logistic map, assuming $r > 0$,

$$x^* = rx^*(1 - x^*) \quad \Longrightarrow \quad x^* = 0 \quad \text{or} \quad x^* = 1 - \frac{1}{r}.$$

The second point is positive only for $r > 1$ and coincides with zero at $r = 1$.

Method. To construct a cobweb, plot $y = f(x)$ and $y = x$. From x_0 on the horizontal axis, go vertically to the curve: the height is x_1. Go horizontally to the diagonal: the horizontal coordinate is now x_1. Repeat vertically to the curve and horizontally to the diagonal.



Cobweb steps: vertical to the map, horizontal to the diagonal.

1.4 Taylor approximation and local stability

For a sufficiently differentiable function near a, a finite Taylor expansion has the form

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + O(h^3).$$

Here $O(h^3)$ denotes an error bounded in magnitude by a constant times $|h|^3$ for sufficiently small h, under appropriate smoothness assumptions. In particular $f(a+h) = f(a) + f'(a)h + o(h)$ for a differentiable f. The formal infinite Taylor series does not automatically equal every smooth function; local approximation is all that is required for linearisation.

Example 1.2 (Taylor expansion of $\log x$ near 2)**.** Since $f'(x) = 1/x$, $f''(x) = -1/x^2$ and $f'''(x) = 2/x^3$,

$$\log x = \log 2 + \frac{x-2}{2} - \frac{(x-2)^2}{8} + \frac{(x-2)^3}{24} + O((x-2)^4).$$

For a first-order approximation, keep only $\log 2 + (x - 2)/2$.

Definition 1.2 (Stability and attraction)**.** An equilibrium is *stable* if arbitrarily small initial perturbations can be kept arbitrarily small for all future times. It is *locally asymptotically stable* if it is stable and nearby states also converge to it. An unstable equilibrium fails stability. These distinctions also apply to continuous-time models.

The sources often use "stable" to mean asymptotically stable. Here a convergence conclusion is stated explicitly.

Write $x_t = x^* + y_t$. Since $f(x^*) = x^*$,

$$y_{t+1} = f(x^* + y_t) - f(x^*) = f'(x^*)y_t + o(y_t).$$

Discarding the smaller terms gives the *linearised* recurrence

$$y_{t+1} = \mu y_t, \qquad y_t = \mu^t y_0, \qquad \mu = f'(x^*).$$

This explicit expression solves the linear approximation, not in general the full nonlinear recurrence.

Proposition 1.3 (Derivative test for a map)**.** *For a continuously differentiable map near x^*:*

- $|f'(x^*)| < 1$ *implies local asymptotic stability;*
- $|f'(x^*)| > 1$ *implies instability;*
- $|f'(x^*)| = 1$ *leaves the linear test inconclusive.*

Explanation. If $|f'(x^*)| < 1$, continuity gives a neighbourhood with $|f'| \le q < 1$. The mean value theorem yields $|f(x)-x^*| \le q|x-x^*|$, so sufficiently nearby iterates remain in the neighbourhood and converge geometrically. If $|f'(x^*)| > 1$, the same argument gives a local expansion factor greater than one, so a nonzero perturbation eventually leaves a fixed small neighbourhood. At equality, higher-order terms may decide the behaviour. □

For $0 < \mu < 1$, the linearised approach keeps the perturbation's sign. For $-1 < \mu < 0$, it alternates sides while shrinking. At $\mu = 0$, higher-order terms determine the detailed approach, but the equilibrium is still locally asymptotically stable.

Example 1.3 (Logistic fixed-point stability)**.** Here $f'(x) = r(1 - 2x)$. At zero, $\mu = r$, so zero attracts for $0 < r < 1$ and is unstable for $r > 1$. At $x^* = 1 - 1/r$,

$$\mu = r\left(1 - 2 + \frac{2}{r}\right) = 2 - r.$$

Thus the positive equilibrium attracts when $1 < r < 3$, and is unstable for $r > 3$. The approach is direct for $1 < r < 2$ and alternating for $2 < r < 3$. The linear test alone does not settle $r = 1$ or $r = 3$.

1.5 Parameters and bifurcations

Definition 1.4 (Bifurcation)**.** A bifurcation is a qualitative change in the dynamics as a parameter varies, for example creation of equilibria or a change in their stability. A bifurcation diagram plots equilibria or periodic states against a parameter; solid and dashed lines often indicate attracting and unstable branches.

Example 1.4 (A parameter-dependent quadratic map)**.** For $x_{t+1} = x_t^2 + x_t - a$, fixed points satisfy $(x^*)^2 = a$. For $a < 0$ there are none; for $a = 0$ there is one; for $a > 0$ there are two: $x^* = \pm\sqrt{a}$. Since $f'(x) = 1 + 2x$,

$$f'(\sqrt{a}) = 1 + 2\sqrt{a} > 1, \qquad f'(-\sqrt{a}) = 1 - 2\sqrt{a}.$$

The positive branch is unstable for every $a > 0$. The negative branch attracts precisely in the range given by the strict derivative test, $|1 - 2\sqrt{a}| < 1$, that is $0 < a < 1$. It is unstable for $a > 1$. At $a = 0$ and $a = 1$, the multiplier equals 1 or -1 respectively, so that test is inconclusive at the parameter value itself. These are the bifurcation values discussed in the sources.

1.6 Period-two cycles and the route to chaos

Definition 1.5 (Period-two cycle)**.** Two distinct values x_-, x_+ form a period-two cycle if $f(x_-) = x_+$ and $f(x_+) = x_-$. They satisfy $f^{\circ 2}(x) = x$, where $f^{\circ 2} = f \circ f$ denotes composition, not squaring the value $f(x)$.

A fixed point of f is also fixed by $f^{\circ 2}$, so those old solutions must be removed when looking for a genuine period-two cycle.

For the logistic map,

$$f^{\circ 2}(x) = r^2 x(1-x)(1-rx(1-x)).$$

Factoring $f^{\circ 2}(x) - x = 0$ gives the known roots $0, 1 - 1/r$ and the additional quadratic

$$x^2 - \left(1 + \frac{1}{r}\right) x + \frac{1}{r} + \frac{1}{r^2} = 0.$$

Its two roots are

$$x_\pm = \frac{r + 1 \pm \sqrt{r^2 - 2r - 3}}{2r}. \tag{1.2}$$

For $r > 0$, they form a genuine distinct positive two-cycle when $r > 3$. At $r = 3$ they merge with the fixed point; for the population interval, we restrict attention to $r \leq 4$.

1.6.1 Stability of a cycle

Let $g = f^{\circ 2}$. By the chain rule, both cycle points have multiplier

$$g'(x_-) = f'(x_+)f'(x_-) = g'(x_+).$$

Using the sum and product of the two roots,

$$\begin{aligned} g'(x_\pm) &= r^2(1 - 2x_-)(1 - 2x_+) \\ &= r^2 \left[1 - 2\left(1 + \frac{1}{r}\right) + 4\left(\frac{1}{r} + \frac{1}{r^2}\right)\right] \\ &= 5 - (r-1)^2. \end{aligned}$$

Hence the cycle attracts for

$$|5 - (r-1)^2| < 1 \quad \Longleftrightarrow \quad 3 < r < 1 + \sqrt{6}.$$

A cycle can be attracting even though neither point is a fixed point of f: we test the product of derivatives over a full cycle.

1.6.2 Period doubling and chaotic behaviour

At $r = 3$, the fixed point loses attraction and the attracting two-cycle appears. At $r = 1 + \sqrt{6} \approx 3.44949$, the cycle multiplier reaches -1 and a four-cycle subsequently attracts. Further doublings give periods $8, 16, \ldots$, accumulating near $r \approx 3.56995$.

Some larger parameters give bounded, aperiodic trajectories sensitive to initial conditions: the source's introductory description of chaos. Periodic windows also occur, including period three near $1 + \sqrt{8} \approx 3.82843$; not every parameter beyond accumulation is chaotic.

Chapter 2

Continuous Mathematical Models

2.1 From a verbal model to an ODE

Continuous time describes a state through a rate of change:

$$\dot{x} = \frac{dx}{dt} = F(x,t), \qquad x(0) = x_0.$$

An equation is *autonomous* when its right-hand side has no explicit time dependence: $\dot{x} = f(x)$. Distinguish this from the discrete equation $x_{t+1} = f(x_t)$: the same expression on the right can give very different behaviour in the two settings.

Method. Choose the state variables and units; state assumptions; identify gains and losses; write rate of change = gains minus losses; check units and physical constraints; impose initial conditions; then solve or analyse equilibria and signs.

For a scalar state of units U, the right side of an ODE must have units U/time. This simple check often catches a missing factor. The models here have smooth right sides, so standard local existence and uniqueness results apply. A solution is followed only while it exists in the relevant domain; not every nonlinear model has solutions for all time.

2.1.1 Partial derivatives and first-order expansion

For $f(x,y)$, partial differentiation varies one argument while fixing the other. In symbols,

$$f_x(x,y) = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h}, \quad f_y(x,y) = \lim_{h\to 0} \frac{f(x,y+h) - f(x,y)}{h}.$$

For instance, $f = x^2 y + \cos y$ gives $f_x = 2xy$ and $f_y = x^2 - \sin y$. For a differentiable function,

$$f(a+X, b+Y) = f(a,b) + f_x(a,b)X + f_y(a,b)Y + o\big(\sqrt{X^2+Y^2}\big).$$

The derivatives evaluated at (a,b) are constants in the linear approximation.

2.2 Scalar population models

2.2.1 Exponential growth

If growth rate is proportional to the population, $\dot{P} = kP$, $k > 0$. The solution is $P(t) = P_0 e^{kt}$, including $P \equiv 0$ when $P_0 = 0$. Positive initial populations increase indefinitely. This can describe a limited time range but ignores eventual resource constraints.

2.2.2 Continuous logistic growth

Competition within a population suggests

$$\dot{P} = kP\left(1 - \frac{P}{K}\right), \qquad k, K > 0. \tag{2.1}$$

Here k has units inverse time and K has population units. The equilibria are 0 and K. For $0 < P < K$, $\dot{P} > 0$; for $P > K$, $\dot{P} < 0$. Thus the phase line points towards K on both sides in the positive domain.

To solve, first treat the equilibria separately, since separation divides by $P(K - P)$. For other initial values,

$$\begin{aligned} \frac{dP}{P(K-P)} &= \frac{k}{K}\,dt, \\ \frac{1}{K}\left(\frac{1}{P} + \frac{1}{K-P}\right)dP &= \frac{k}{K}\,dt, \\ \log\left|\frac{P}{K-P}\right| - \log\left|\frac{P_0}{K-P_0}\right| &= kt. \end{aligned}$$

The sign of $P/(K-P)$ is preserved along a non-equilibrium trajectory. After applying the initial condition and rearranging,

$$P(t) = \frac{KP_0}{P_0 + (K - P_0)e^{-kt}}. \tag{2.2}$$

This formula also gives both equilibrium solutions when $P_0 = 0$ or K. For $P_0 > 0$, its denominator remains positive for $t \geq 0$ and $P(t) \to K$. If $P_0 > K$, absolute values in the logarithmic derivation are essential, even though the final formula has the same form.

2.3 Phase lines and scalar stability

For $\dot{P} = f(P)$, an equilibrium satisfies $f(P^*) = 0$. Write $P = P^* + p$ and expand:

$$\dot{p} = f(P^* + p) = f'(P^*)p + o(p).$$

The linearised equation $\dot{p} = \lambda p$ has solution $p(t) = p_0 e^{\lambda t}$, with $\lambda = f'(P^*)$. For a continuously differentiable f near P^*:

- $f'(P^*) < 0$ implies local asymptotic stability;
- $f'(P^*) > 0$ implies instability;
- $f'(P^*) = 0$ leaves the linear test inconclusive.

The condition here is a *negative derivative*, not derivative modulus less than 1. The latter belongs to discrete maps.

Example 2.1 (A periodic array of equilibria: source Example 2.1)**.** For $\dot{P} = \sin P$, equilibria satisfy $P^* = n\pi$, $n \in \mathbb{Z}$. Since $f'(P^*) = \cos(n\pi) = (-1)^n$, odd multiples attract and even multiples are unstable. Between neighbouring even multiples $2j\pi$ and $2(j+1)\pi$, every non-equilibrium solution tends to $(2j+1)\pi$. If P denotes a population, restrict the mathematical phase line to nonnegative values.

Example 2.2 (Additional handwritten example, 15 October)**.** Consider $\dot{x} = f(x) = e^{-x}\sin x$ for initial values $x_0 \in [0, 6]$. The equilibria in this interval are 0 and π. Since $f'(x) = e^{-x}(\cos x - \sin x)$,

$$f'(0) = 1 > 0, \qquad f'(\pi) = -e^{-\pi} < 0.$$

Thus 0 is unstable, while π attracts. For $0 < x < \pi$ the state moves right; for $\pi < x \leq 6 < 2\pi$ it moves left. Every $x_0 \in (0, 6]$ tends to π, while $x_0 = 0$ stays fixed.

To interpret the changing speed, note $|\dot{x}| = |f(x)|$. The positive velocity has its maximum at $x = \pi/4$; the negative velocity has its minimum at $x = 5\pi/4$. A right-moving state below $\pi/4$ first speeds up and then slows down. A left-moving state above $5\pi/4$ first speeds up as it approaches $5\pi/4$, then slows towards π. A negative velocity and a decreasing speed are different statements.

2.4 A mixing model

A tank initially contains 100 litres of pure water. Brine with concentration 5 grams per litre enters at 2 litres per minute; an equal outflow keeps the volume constant. Assume perfect instantaneous mixing. Let $M(t)$ be the salt mass in grams and t be measured in minutes.

$$\text{inflow salt rate} = 5 \times 2 = 10 \text{ g/min}, \qquad \text{outflow salt rate} = \frac{M}{100} \times 2 = \frac{M}{50} \text{ g/min}.$$

Therefore $\dot{M} = 10 - M/50$, with $M(0) = 0$. An integrating factor gives

$$\begin{aligned} \frac{d}{dt}\left(Me^{t/50}\right) &= 10e^{t/50}, \\ M(t) &= 500 + Ce^{-t/50}, \\ M(0) = 0 \quad &\Longrightarrow \quad C = -500. \end{aligned}$$

The mass and concentration are

$$M(t) = 500(1 - e^{-t/50}) \text{ g}, \qquad c(t) = \frac{M(t)}{100} = 5(1 - e^{-t/50}) \text{ g/L}.$$

As $t \to \infty$, the tank concentration approaches the incoming concentration. If inflow and outflow volumes differed, the volume would vary and the outflow concentration would be $M(t)/V(t)$ instead.

2.5 Coupled systems, phase planes and nullclines

A two-species model has the form

$$\dot{x} = f(x, y), \qquad \dot{y} = g(x, y).$$

Terms proportional to xy represent encounters under a well-mixed assumption. Their signs encode the interaction. For positive constants:

Interaction	$\dot{x}$	$\dot{y}$
Predation (x prey, y predator)	$ax - bxy$	$-cy + dxy$
Competition	$ax - bxy$	$cy - dxy$
Mutual benefit	$ax + bxy$	$cy + dxy$

These are simplified models; sign information alone does not capture all resource limitations or guarantee bounded populations.

Definition 2.1 (Phase portrait and nullclines)**.** A trajectory is the curve traced by $(x(t), y(t))$ in the phase plane, with arrows showing increasing time. The vector field is $(f(x, y), g(x, y))$. The x-nullclines satisfy $f(x, y) = 0$, where nonzero arrows are vertical. The y-nullclines satisfy $g(x, y) = 0$, where nonzero arrows are horizontal. Equilibria satisfy both equations simultaneously.

Under uniqueness, two distinct trajectories cannot cross at a non-equilibrium point: the same state cannot have two different future solutions. A nullcline is generally not a trajectory, and a point on just one nullcline need not be an equilibrium.

Example 2.3 (Reading the source phase portrait: source Example 2.3)**.** The supplied sketch has rightward flow on the x-axis and downward flow on the y-axis, with an unstable origin, an unstable interior equilibrium, and a closed orbit attracting nearby trajectories. It illustrates an attracting *limit cycle*: an isolated periodic orbit. One cannot infer this entire portrait from interaction signs alone. In particular, the closed Lotka–Volterra orbits studied below are not attracting limit cycles, because they form a family of neighbouring cycles.

Method. For a phase portrait: solve $f = g = 0$; draw both types of nullcline; check signs of f and g in each region; combine them into flow arrows; then draw trajectories consistent with those directions and uniqueness.

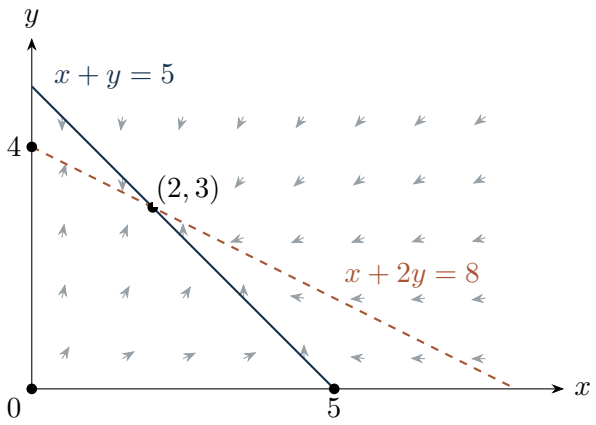
Example 2.4 (Competing species: source Section 2.4.2)**.** Consider

$$\dot{x} = x(5 - x - y), \qquad \dot{y} = y(8 - x - 2y).$$

The x-nullclines are $x = 0$ and $x + y = 5$; the y-nullclines are $y = 0$ and $x + 2y = 8$. Their intersections give

$$(0, 0), \qquad (5, 0), \qquad (0, 4), \qquad (2, 3).$$

For $x, y > 0$, $\dot{x}$ has the sign of $5 - x - y$, and $\dot{y}$ has the sign of $8 - x - 2y$. At $(1, 1)$, both are positive, so arrows point up and right. Above both oblique nullclines, both are negative, so arrows point down and left. The axes are invariant because the corresponding population derivative vanishes.



Competition model: oblique nullclines and normalised flow arrows.

2.6 Linearisation of two-dimensional systems

At an equilibrium (x^*, y^*), write $x = x^* + X$ and $y = y^* + Y$. First-order expansion gives the linearised system

$$\begin{pmatrix} \dot{X} \\ \dot{Y} \end{pmatrix} = J \begin{pmatrix} X \\ Y \end{pmatrix}, \qquad J = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}\Bigg|_{(x^*,y^*)}.$$

The matrix J is the *Jacobian at the equilibrium.* For the original nonlinear system there are also higher-order remainder terms.

Try a mode $(X, Y)^\mathsf{T} = \mathbf{v}e^{\lambda t}$, with $\mathbf{v} \neq \mathbf{0}$. Substitution gives $J\mathbf{v} = \lambda\mathbf{v}$. Thus λ is an eigenvalue and must satisfy

$$\det(\lambda I - J) = (\lambda - A)(\lambda - D) - BC = 0. \tag{2.3}$$

This determinant approach works even when $B = 0$ or $C = 0$; dividing by a matrix entry could lose those cases.

For distinct eigenvalues with independent eigenvectors, the general linear solution is

$$\begin{pmatrix} X \\ Y \end{pmatrix} = c_1\mathbf{v}_1e^{\lambda_1 t} + c_2\mathbf{v}_2e^{\lambda_2 t}.$$

There are two independent constants, not four unrelated coefficients in X and Y. Repeated eigenvalues may require a term involving $te^{\lambda t}$. Complex conjugate modes combine into real sine and cosine terms. Superposition applies to the homogeneous linearised system, not generally to the original nonlinear model.

Proposition 2.2 (Local linear stability)**.** *For a continuously differentiable nonlinear vector field, if all eigenvalues of its equilibrium Jacobian have negative real part, the equilibrium is locally asymptotically stable. If at least one has positive real part, it is unstable. If some real parts are zero and none are positive, linearisation alone generally does not decide nonlinear stability.*

These standard linearisation results justify using the modes as a local stability test. The linear solution is not an exact formula for the full nonlinear trajectory, even when it predicts stability correctly.

2.6.1 Trace and determinant: a supplementary shortcut

Let $\tau = A + D$ and $\Delta = AD - BC$. The characteristic equation is $\lambda^2 - \tau\lambda + \Delta = 0$.

- $\Delta < 0$: one positive and one negative eigenvalue, an unstable saddle.
- $\Delta > 0$, $\tau < 0$: negative real parts, local attraction.
- $\Delta > 0$, $\tau > 0$: positive real parts, instability.
- $\Delta > 0$, $\tau = 0$: purely imaginary eigenvalues; no nonlinear stability conclusion from the linear test alone.

For $\Delta > 0$, the sign of $\tau^2 - 4\Delta$ distinguishes two real modes from complex conjugate modes. The cases $\Delta = 0$ require further analysis.

2.6.2 Return to the competition example

At $(2,3)$, direct substitution gives

$$\dot{X} = -2X - 2Y - X^2 - XY,$$
$$\dot{Y} = -3X - 6Y - XY - 2Y^2.$$

Dropping the quadratic terms gives

$$J = \begin{pmatrix} -2 & -2 \\ -3 & -6 \end{pmatrix}, \qquad \lambda^2 + 8\lambda + 6 = 0, \qquad \lambda = -4 \pm \sqrt{10}.$$

Both eigenvalues are negative, so coexistence is locally asymptotically stable. For reference, the general Jacobian is

$$J(x,y) = \begin{pmatrix} 5 - 2x - y & -x \\ -y & 8 - x - 4y \end{pmatrix}.$$

The other equilibria have eigenvalues $5, 8$ at $(0,0)$; $-5, 3$ at $(5,0)$; and $1, -8$ at $(0,4)$. Thus the origin is repelling and the two nonzero axis equilibria are saddles. These extra classifications complete the local picture without asserting global attraction for every initial state.

2.7 The Lotka–Volterra predator–prey model

Convention for this section: x is predator population, y is prey. This reverses the labels in the earlier illustrative interaction table. With $a, b, c, d > 0$,

$$\dot{x} = x(-a + by), \qquad \dot{y} = y(c - dx). \tag{2.4}$$

Predators decline without prey, prey grow without predators, and encounters benefit predators while reducing prey. Positive populations stay positive: for example $x(t) = x_0 \exp\left(\int_0^t [-a + by(s)]\, ds\right)$. The axes are invariant and are analysed separately from the positive interior.

The nullclines are $x = 0$, $y = a/b$ for $\dot{x} = 0$, and $y = 0$, $x = c/d$ for $\dot{y} = 0$. Equilibria are $(0,0)$ and $(c/d, a/b)$. Above $y = a/b$ predators grow; to the right of $x = c/d$ prey decrease. For this predator-on-horizontal-axis convention, the positive orbits run clockwise around coexistence.

2.7.1 Harvesting

Per-capita harvesting rates $h_1, h_2 \geq 0$ give

$$\dot{x} = x(-a - h_1 + by), \qquad \dot{y} = y(c - h_2 - dx).$$

If $c > h_2$, the positive equilibrium shifts to

$$\left(\frac{c - h_2}{d}, \frac{a + h_1}{b}\right).$$

Thus its predator value falls and prey value rises relative to the original coexistence equilibrium. These are equilibrium or cycle-average comparisons, not a claim that every instantaneous prey population rises. If $c \leq h_2$, there is no positive coexistence equilibrium of this form.

2.7.2 Local analysis

At the origin, the linearised equations are $\dot{X} = -aX$, $\dot{Y} = cY$, so the origin is a saddle. At coexistence,

$$J = \begin{pmatrix} 0 & bc/d \\ -ad/b & 0 \end{pmatrix}, \qquad \lambda^2 = -ac, \qquad \lambda = \pm i\sqrt{ac}.$$

Let $\omega = \sqrt{ac}$ and $B = bc/d$. The real linear solution can be written

$$X(t) = A_1 \cos(\omega t) + A_2 \sin(\omega t),$$
$$Y(t) = \frac{\omega}{B}[-A_1 \sin(\omega t) + A_2 \cos(\omega t)].$$

It oscillates with period $2\pi/\sqrt{ac}$, using two independent constants. This is the small-amplitude linearised period, not the exact period of every nonlinear orbit. Purely imaginary eigenvalues alone do not prove nonlinear stability; we next use a conserved quantity.

2.7.3 A first integral and closed trajectories

For $x, y > 0$ and away from a vertical trajectory tangent, take the ratio:

$$\frac{dy}{dx} = \frac{y(c - dx)}{x(-a + by)}.$$

Separation yields

$$\int \left(-\frac{a}{y} + b\right) dy = \int \left(\frac{c}{x} - d\right) dx,$$

so a trajectory satisfies

$$c \log x - dx + a \log y - by = \text{constant}. \tag{2.5}$$

Logs require *strictly positive* populations, not merely nonnegative ones. Direct differentiation verifies the conserved expression along the whole positive trajectory, including points where the ratio dy/dx was unavailable. Exponentiating gives

$$F(x)G(y) = H > 0, \qquad F(x) = x^c e^{-dx}, \quad G(y) = y^a e^{-by}.$$

Each factor tends to zero at both ends of its positive domain. Also

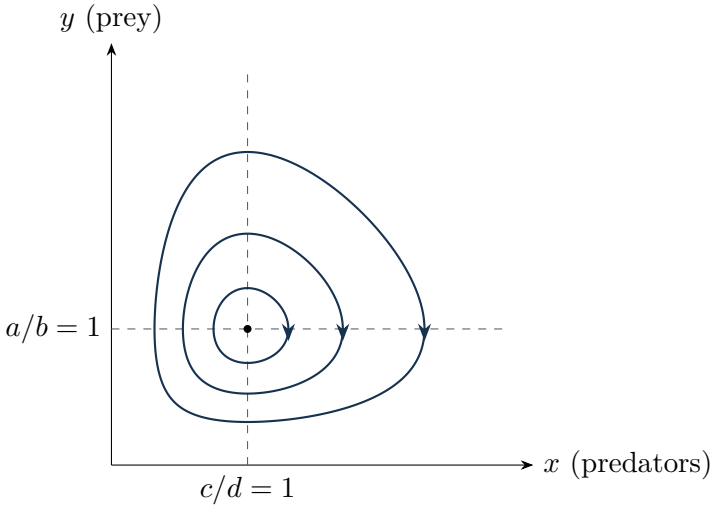
$$F'(x) = x^{c-1} e^{-dx}(c - dx), \qquad G'(y) = y^{a-1} e^{-by}(a - by).$$

Thus F has a unique maximum $M_1 = (c/d)^c e^{-c}$ at c/d; G has a unique maximum $M_2 = (a/b)^a e^{-a}$ at a/b. At $H = M_1 M_2$, the level set is just the equilibrium. For $0 < H < M_1 M_2$, horizontal and vertical lines through the equilibrium each meet the level set twice, and the trajectory is a closed curve surrounding it.

Supplementary justification. Equivalently,

$$V(x, y) = dx - c \log x + by - a \log y$$

is strictly convex, tends to $+\infty$ at the boundary and at infinity, and has its unique minimum at coexistence. Its non-minimal level sets are compact smooth closed curves with no equilibrium on them. The nonvanishing vector field follows each such curve periodically. It also follows that coexistence is stable in the non-attracting sense: nearby level curves remain nearby, but no other one converges to the centre. It is therefore a nonlinear *centre*, not an asymptotically stable state and not an attracting limit cycle.



Closed orbits around the centre $(1, 1)$, for $a = b = c = d = 1$.

2.7.4 Average populations over a cycle

For a periodic positive orbit of period T,

$$\frac{\dot{y}}{y} = c - dx \quad \Longrightarrow \quad \frac{1}{T}\int_0^T x(t)\,dt = \frac{c}{d} - \frac{\log y(T) - \log y(0)}{dT} = \frac{c}{d}.$$

Similarly, integrating $\dot{x}/x = -a + by$ gives

$$\overline{x} = \frac{c}{d}, \qquad \overline{y} = \frac{a}{b}.$$

The time averages equal the equilibrium populations although the instantaneous values oscillate. For the harvested model with positive coexistence, the same calculation uses $a + h_1$ and $c - h_2$.

2.8 The SIR epidemic model

This is a mathematical compartment model under idealised assumptions, not a fitted description of a particular current epidemic. Let S denote susceptible individuals, I infectious individuals, and R removed individuals (recovered, isolated or dead). Assume a closed, well-mixed population, no demographic turnover or return to susceptibility, and transitions $S \to I \to R$. The model is

$$\dot{S} = -\alpha SI, \qquad \dot{I} = \alpha SI - \beta I, \qquad \dot{R} = \beta I, \qquad \alpha, \beta > 0. \tag{2.6}$$

With population counts as variables, α has units $(\text{population} \times \text{time})^{-1}$ and β has units inverse time. Some other conventions use an infection term divided by population size; we consistently follow the supplied equations without that divisor.

Take $S(0) = S_0 > 0$, $I(0) = I_0 > 0$ and $R(0) = 0$. Adding equations gives $S+I+R = N = S_0+I_0$, a conserved total. Thus $R = N - S - I$ and the physical S–I phase region is the triangle

$$S \geq 0, \qquad I \geq 0, \qquad S + I \leq N.$$

The full line $I = 0$ consists of equilibria. The S-nullclines are $S = 0$ and $I = 0$; the I-nullclines are $I = 0$ and $S = \rho$, where

$$\rho = \frac{\beta}{\alpha}.$$

Inside the positive triangle, $\dot{S} < 0$, so every trajectory moves left. If $S > \rho$, $\dot{I} > 0$; if $S < \rho$, $\dot{I} < 0$. The dimensionless initial growth threshold is $\alpha S_0/\beta = S_0/\rho$.

2.8.1 An exact phase-plane relation

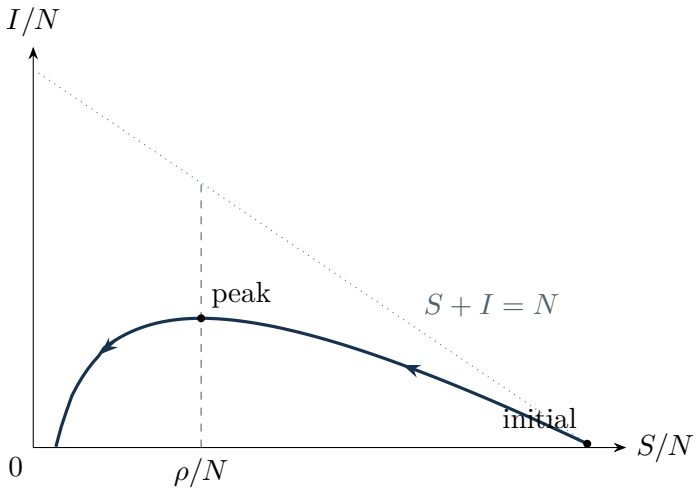
For $S, I > 0$,

$$\frac{dI}{dS} = -1 + \frac{\rho}{S}.$$

Integrating and applying initial conditions gives

$$I(S) = N - S + \rho \log \frac{S}{S_0}. \tag{2.7}$$

This relates the two populations without explicitly solving for time.



Illustrative SIR trajectory; arrows move towards decreasing S.

2.8.2 When does the infection peak?

If $S_0 > \rho$, I initially rises. Its subsequent peak occurs when $S = \rho$, giving

$$I_{\max} = N - \rho + \rho \log \frac{\rho}{S_0}.$$

If $S_0 \leq \rho$, the trajectory starts on or to the left of this threshold, so its maximum for $t \geq 0$ is I_0, not the value of an unreachable turning point obtained by extending the phase curve backwards. The peak formula therefore needs the outbreak condition $S_0 > \rho$.

2.8.3 Removed population and final size

Taking the ratio of the S and R equations gives

$$\frac{dS}{dR} = -\frac{S}{\rho} \quad \Longrightarrow \quad S(R) = S_0 e^{-R/\rho}.$$

Since $0 \leq R \leq N$, $S(t) \geq S_0 e^{-N/\rho} > 0$. Susceptibles are not exhausted completely; infections stop growing when their reproduction rate falls below removals. Substitution into $\dot{R} = \beta I$ gives the exact scalar equation

$$\dot{R} = \beta\big[N - R - S_0 e^{-R/\rho}\big], \qquad R(0) = 0. \tag{2.8}$$

It is suitable for numerical integration but does not generally have a simple elementary explicit solution.

Supplementary final-size relation. Since S decreases and R increases while both remain bounded, they have limits S_∞, R_∞. Then $I = N - S - R$ has a limit too. That limit must be zero, since $\dot{R} = \beta I$ and a positive limiting I would force unbounded R. Therefore

$$S_\infty = S_0 e^{-(N-S_\infty)/\rho}, \qquad R_\infty = N - S_\infty.$$

This implicit equation determines the physical final state reached from the given initial conditions; it is not an instruction to choose an arbitrary algebraic root.

2.9 An approximate explicit SIR solution

The sources obtain a solvable approximation by assuming $R/\rho \ll 1$. Expanding the exponential in (2.8) gives

$$e^{-R/\rho} = 1 - \frac{R}{\rho} + \frac{R^2}{2\rho^2} + O\big((R/\rho)^3\big).$$

With $I_0 = N - S_0$, define $q = S_0/\rho - 1$ and $a = S_0/(2\rho^2)$. The quadratic approximation is

$$\dot{R}_{\text{app}} = \beta(I_0 + qR_{\text{app}} - aR_{\text{app}}^2), \qquad R_{\text{app}}(0) = 0. \tag{2.9}$$

Using a separate subscript emphasises that this is a different differential equation, approximating the exact model only where the expansion is accurate.

2.9.1 Solving the quadratic equation

The source marks the detailed integration non-examinable. Retain it as an optional worked derivation. Put

$$\kappa = \sqrt{q^2 + 4aI_0}, \qquad b = \frac{q}{2a}, \qquad A = \frac{\kappa}{2a}.$$

Since $I_0 > 0$, $\kappa > |q|$ and hence $|q/\kappa| < 1$. Completing the square gives

$$I_0 + qR - aR^2 = a\big[A^2 - (R-b)^2\big].$$

Let $z = (R-b)/A$. Then

$$\dot{z} = \frac{\beta\kappa}{2}(1 - z^2), \qquad \operatorname{artanh} z = \frac{\beta\kappa t}{2} + C.$$

At $t = 0$, $z = -q/\kappa$, so set $\phi = \operatorname{artanh}(q/\kappa)$ to obtain

$$R_{\text{app}}(t) = \frac{\rho^2}{S_0}\left[q + \kappa \tanh\left(\frac{\beta\kappa t}{2} - \phi\right)\right]. \tag{2.10}$$

Here artanh is the inverse hyperbolic tangent, not the reciprocal of tanh. Differentiating the approximate solution gives

$$\dot{R}_{\text{app}}(t) = \frac{\beta\kappa^2\rho^2}{2S_0} \operatorname{sech}^2\left(\frac{\beta\kappa t}{2} - \phi\right).$$

The initial condition is a useful check: $\tanh(-\phi) = -q/\kappa$, so $R_{\text{app}}(0) = 0$.

Remark (Keep the approximation consistent). For the exact model, $S = S_0 e^{-R/\rho}$ and $I = N - R - S$. Inserting R_{app} into these expressions gives approximate reconstructions, not an exact SIR solution. Alternatively $I_{\text{app}} = \dot{R}_{\text{app}}/\beta$ follows from the quadratic truncation. These two reconstructions agree only to the retained order: they should not be described as identical exact functions. A large numerical ρ is not sufficient by itself; the dimensionless ratio R/ρ must remain small over the time range of interest.

2.9.2 What the source's historical illustrations show

For the supplied boarding-school illustration, the stated inputs are $N = 760$, $I_0 = 1$, $S_0 = 759$ and a peak near 300 infections. Using the exact phase-plane peak formula gives an equation for ρ:

$$760 - \rho + \rho \log(\rho/759) = 300.$$

On $0 < \rho < 759$, numerical solution gives roughly $\rho \approx 195.0$. The source rounds this to about 200. The peak height determines this parameter ratio, but not the time scale: one needs timing information to identify β, then $\alpha = \beta/\rho$.

For the source's rounded choice $\rho = 200$, $q \approx 2.795$, $\kappa \approx 2.802$, and $q/\kappa \approx 0.998$. Thus ϕ is around 3.4, not artanh(0.014) as printed. Also the total number removed can be much larger than 200, so the small-R/ρ approximation cannot be assumed valid over the whole outbreak. These are model illustrations from the handout; the underlying data are not supplied as a table, and no data fit has been reproduced here.

The Bombay plague illustration similarly compares a curve proportional to $\dot{R}$, rather than cumulative R, with deaths per unit time. Interpreting deaths as a fixed fraction of removals requires an extra observation assumption; the compartment equations alone do not provide that conversion. The source's plotted observations and fitted parameters are not reconstructed as new evidence.

2.10 A second-order equation as a planar system

For

$$\ddot{x} + 3\dot{x} + 2\sin x = 0,$$

set $y = \dot{x}$. Then

$$\dot{x} = y, \qquad \dot{y} = -3y - 2\sin x.$$

The nullclines are $y = 0$ and $y = -(2/3)\sin x$, and the equilibria are $(n\pi, 0)$ for integers n. At such a point,

$$J = \begin{pmatrix} 0 & 1 \\ -2\cos(n\pi) & -3 \end{pmatrix}, \qquad \lambda^2 + 3\lambda + 2\cos(n\pi) = 0.$$

For even n, the eigenvalues are $-1, -2$, giving local attraction. For odd n, they are $(-3 \pm \sqrt{17})/2$, with opposite signs, giving an unstable saddle. This conversion lets the same equilibrium and Jacobian methods analyse second-order motion, connecting this chapter with mechanics.

Method. For a continuous model: find zero rates, not intersections with $y = x$. Use derivative signs in one dimension and Jacobian eigenvalue real parts in two dimensions. If the linear test has zero real parts, look for nonlinear information such as a first integral. State any domain or small-parameter assumptions before using an explicit approximation.

Chapter 3

Mechanics: Simple Forces

3.1 Newton's laws and inertial frames

The models in this chapter treat bodies as point particles of constant mass. Position, velocity, acceleration and force are vectors; in one-dimensional motion we use signed scalar components along a chosen positive axis.

Definition 3.1 (Inertial frame)**.** An inertial frame is a reference frame in which a force-free particle moves with constant velocity. Newton's laws below are used in such a frame.

1. **First law.** An isolated particle moves in a straight line at constant velocity. Zero velocity is just one possibility.
2. **Second law.** For fixed mass m, the net force is $\mathbf{F}_{\text{net}} = m\mathbf{a} = m\ddot{\mathbf{r}}$.
3. **Third law.** The force on particle 1 due to particle 2 equals minus the force on particle 2 due to particle 1: $\mathbf{F}_{1\leftarrow 2} = -\mathbf{F}_{2\leftarrow 1}$. The strong form used for central interactions also places them along the line joining the particles.

Third-law force pairs act on different bodies: they do not cancel on a free-body diagram of just one body. Multiple forces acting on the same body must be summed before applying the second law. In an accelerating frame, extra inertial-force terms are needed; one cannot simply reuse the same net-force equation unchanged.

3.2 Balancing forces

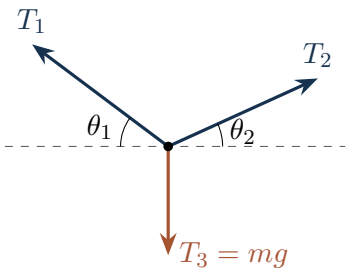
Near Earth's surface, weight has magnitude mg, directed downwards, with $g \approx 9.8\ \mathrm{m\,s^{-2}}$. Mass is measured in kilograms, force in newtons, and $1\ \mathrm{N} = 1\ \mathrm{kg\,m\,s^{-2}}$. Weight depends on the gravitational field; mass does not disappear when an object is weightless.

A taut light string exerts a tension along itself, pulling on the body to which it is attached. A smooth contact surface exerts a normal reaction perpendicular to the surface. For a light string over an ideal smooth massless pulley, the tension magnitude is the same on both sides. A string cannot push; if a calculation requires negative tension, the taut string model must be reconsidered.

Static equilibrium requires zero acceleration and zero initial velocity. Zero net force alone gives constant velocity, which need not be rest. To balance forces on a particle, resolve the vector equation $\sum \mathbf{F} = \mathbf{0}$ into convenient perpendicular directions.

Method. Draw a separate free-body diagram for each body or junction. Choose axes and mark the angles from those axes. Resolve every force, then apply $\sum F = ma$ along each axis. Do not add a body's own third-law reaction force to its own diagram.

Example 3.1 (Three supporting cables: source Example 3.1)**.** A mass m hangs from a vertical cable, attached at a light junction to two inclined cables making angles θ_1, θ_2 above the horizontal. Let the inclined tensions be T_1, T_2 and the vertical one T_3.



Free-body diagram of the light junction; angles are from horizontal.

For the mass, equilibrium gives $T_3 = mg$. For the junction, resolve horizontally and vertically:

$$T_1 \cos\theta_1 = T_2 \cos\theta_2, \qquad T_1 \sin\theta_1 + T_2 \sin\theta_2 = mg.$$

Eliminating one tension and using the sine addition identity gives

$$T_1 = \frac{mg\cos\theta_2}{\sin(\theta_1+\theta_2)}, \qquad T_2 = \frac{mg\cos\theta_1}{\sin(\theta_1+\theta_2)}, \qquad T_3 = mg.$$

For equal angles θ, both inclined tensions equal $mg/(2\sin\theta)$. Nearly horizontal cables need very large tension, consistent with the small vertical components of their forces.

Example 3.2 (Blocks on smooth inclines: source Example 3.2)**.** Two masses $m_1 = 1$ kg and $m_2 = \sqrt{2}$ kg are joined by a taut light string over an ideal pulley. The smooth inclines have angles $\pi/4$ and θ above horizontal. For equilibrium, resolve along each plane:

$$T = m_1 g \sin(\pi/4) = \frac{g}{\sqrt{2}}, \qquad T = m_2 g \sin\theta = \sqrt{2} g \sin\theta.$$

Thus $\sin\theta = 1/2$; with $0 < \theta < \pi/2$, $\theta = \pi/6$. Perpendicular resolution gives

$$R_1 = m_1 g\cos(\pi/4) = \frac{g}{\sqrt{2}}, \qquad R_2 = m_2 g\cos(\pi/6) = g\sqrt{\frac{3}{2}}.$$

With masses substituted numerically in kilograms, these force values are in newtons. The reaction is not generally mg: only the perpendicular component of weight is balanced by it.

3.3 Hooke's law and connected springs

A spring of natural length ℓ and stiffness $k > 0$ exerts a restoring force proportional to signed extension e:

$$\text{spring tension} = ke, \qquad \text{force on a displaced end opposes } e.$$

Stiffness has units N/m. Positive extension means stretching; negative extension means compression, when a spring pushes outwards. Hooke's law is an approximation within the spring's elastic range. An elastic string differs: it supplies tension when extended but goes slack when shorter than its natural length.

Example 3.3 (Three vertical springs: source Example 3.3)**.** Three identical light springs of natural length ℓ and stiffness k connect a ceiling to a floor 3ℓ below it. A mass m is attached at each of the two intermediate junctions. Measure distance downwards from the ceiling. Let the mass positions be $\ell + x_1$ and $2\ell + x_2$. The signed spring extensions are

$$e_1 = x_1, \qquad e_2 = x_2 - x_1, \qquad e_3 = -x_2.$$

For the upper mass, downward forces minus upward forces give

$$mg + k(x_2 - x_1) - kx_1 = 0.$$

For the lower mass, the middle spring force is upward with signed magnitude $k(x_2 - x_1)$, and the bottom spring contributes downward signed force $-kx_2$. Thus

$$mg - k(x_2 - x_1) - kx_2 = 0.$$

The two equations become

$$2x_1 - x_2 = \frac{mg}{k}, \qquad 2x_2 - x_1 = \frac{mg}{k}.$$

Subtracting gives $x_1 = x_2$, and hence both equal mg/k. The mass positions below the ceiling are

$$\ell + \frac{mg}{k}, \qquad 2\ell + \frac{mg}{k}.$$

The spring lengths are $\ell + mg/k$, ℓ, and $\ell - mg/k$. The bottom spring is compressed and pushes upwards on the lower mass; the middle spring has zero extension. The ideal geometry requires $mg/k < \ell$, and physical validity also requires staying in the elastic range.

3.4 Equations of one-dimensional motion

Choose a signed coordinate $x(t)$ and write

$$v = \dot{x}, \qquad a = \ddot{x} = \frac{dv}{dt}, \qquad m\ddot{x} = F(t, x, \dot{x}).$$

Where velocity can be considered locally as a function of position, the chain rule gives a third useful acceleration form:

$$a = \frac{dv}{dt} = \frac{dv}{dx}\frac{dx}{dt} = v\frac{dv}{dx}.$$

Use dv/dt for a velocity–time relation and $v\,dv/dx$ for a velocity–position relation. Turning points may require treating separate branches of $v(x)$; one cannot divide by v indiscriminately at $v = 0$.

3.5 Constant force and time-dependent force

3.5.1 Constant acceleration

For $\dot{v} = a$ constant, with $x(0) = x_0$ and $v(0) = u$, integrate twice:

$$v(t) = u + at, \qquad x(t) = x_0 + ut + \frac{1}{2}at^2.$$

Alternatively, integrating $v\,dv/dx = a$ gives

$$v^2 = u^2 + 2a(x - x_0).$$

These formulae require constant acceleration and consistent signed directions.

Example 3.4 (A sinusoidal force: source Example 3.4)**.** Suppose $F(t) = mA\sin(\omega t)$, with $\omega > 0$, $x(0) = 0$ and $v(0) = v_0$. Here A is an acceleration amplitude, not a force. The equation $\ddot{x} = A\sin(\omega t)$ integrates to

$$v(t) = v_0 + \frac{A}{\omega}(1 - \cos(\omega t)),$$
$$x(t) = \left(v_0 + \frac{A}{\omega}\right) t - \frac{A}{\omega^2}\sin(\omega t).$$

The sine term stays bounded, while the first term is linear in t. Consequently $x(t) \to +\infty$ exactly when $v_0 > -A/\omega$; it tends to $-\infty$ for $v_0 < -A/\omega$, and remains bounded at equality. Divergence of position to $+\infty$ does not necessarily mean the velocity is positive at every sufficiently late instant: bounded oscillations can remain on top of the drift.

3.6 Quadratic resistance: direction matters

A drag force must oppose velocity. With constant $k > 0$, quadratic drag per unit mass is represented in one dimension by $-kv|v|$. Choose upward displacement y and upward signed velocity $v = \dot{y}$. With gravity, a single signed equation is

$$\dot{v} = -g - kv|v|.$$

During ascent ($v > 0$), it becomes $\dot{v} = -g - kv^2$. During descent ($v < 0$), it becomes $\dot{v} = -g + kv^2$. Using $-kv^2$ for both upward and downward velocities would make drag point downward in both cases, which is physically wrong. Here k has units inverse length.

Example 3.5 (Maximum height: source Example 3.5)**.** Launch from $y = 0$ with $v(0) = u > 0$. At the highest point $y = h$, $v = 0$. For the ascending part use $v\,dv/dy = -g - kv^2$, so

$$\begin{aligned} h &= -\int_u^0 \frac{v\,dv}{g + kv^2} \\ &= \frac{1}{2k}\log\left(\frac{g + ku^2}{g}\right) = \frac{1}{2k}\log\left(1 + \frac{ku^2}{g}\right). \end{aligned}$$

As $k \to 0$, $\log(1 + z) \sim z$ recovers the drag-free height $u^2/(2g)$. Only ascent was used; the drag sign changes after the turning point.

Example 3.6 (Downward terminal speed: source Example 3.6)**.** Now choose displacement positive *downwards* and let $v(0) = v_0 \geq 0$. Gravity is positive and drag negative, so

$$\dot{v} = g - kv^2 = k(V^2 - v^2), \qquad V = \sqrt{\frac{g}{k}}.$$

The equilibrium $v = V$ is the terminal speed. If $v_0 = V$, the speed stays constant. Otherwise, separation gives

$$\int_{v_0}^{v(t)} \frac{dw}{V^2 - w^2} = kt, \qquad \frac{1}{V^2 - w^2} = \frac{1}{2V}\left(\frac{1}{V - w} + \frac{1}{V + w}\right).$$

Hence

$$\log\left|\frac{V + v}{V - v}\frac{V - v_0}{V + v_0}\right| = 2Vkt.$$

A solution cannot cross the equilibrium V by uniqueness, so $V - v$ keeps the sign of $V - v_0$. Rearranging gives the expression valid for all $v_0 \geq 0$:

$$v(t) = V\frac{(V + v_0)e^{2Vkt} - (V - v_0)}{(V + v_0)e^{2Vkt} + (V - v_0)}. \tag{3.1}$$

It also includes $v_0 = V$ by direct substitution. For $v_0 < V$ the speed increases towards V; for $v_0 > V$ it decreases towards V. Equivalently, when $v_0 < V$,

$$v(t) = V \tanh\left(Vkt + \operatorname{artanh}(v_0/V)\right).$$

For $v_0 > V$, the corresponding hyperbolic form uses coth, not a real $\operatorname{artanh}(v_0/V)$. The exponential fraction avoids this case split.

3.7 Linear resistance and terminal velocity

For constant applied force per unit mass a_0 and linear drag coefficient $\lambda > 0$, the signed equation is

$$\dot{v} = a_0 - \lambda v.$$

The drag term $-\lambda v$ automatically reverses with the velocity. The terminal velocity is $v_\infty = a_0/\lambda$, which may be positive, negative or zero. The coefficient λ has units inverse time. An integrating factor gives, for every real v_0,

$$v(t) = v_\infty + (v_0 - v_\infty)e^{-\lambda t},$$
$$x(t) = x_0 + v_\infty t + \frac{v_0 - v_\infty}{\lambda}(1 - e^{-\lambda t}).$$

The difference from terminal velocity preserves its sign and decays exponentially. Starting exactly at terminal velocity produces uniform motion.

Example 3.7 (A bullet in a block: source Example 3.8)**.** A particle enters a resistive block at $x = 0$ with speed $u > 0$. Resistance per unit mass is kv, with $k > 0$, and gravity is neglected. Then $\dot{v} = -kv$ and

$$v(t) = ue^{-kt}, \qquad x(t) = \frac{u}{k}(1 - e^{-kt}).$$

The limiting penetration is u/k. It is approached as $t \to \infty$; in this ideal model, the particle does not stop exactly in finite time. A block of width at least u/k prevents finite-time escape through the far side. To reach half the limiting distance,

$$\frac{u}{2k} = \frac{u}{k}(1 - e^{-kT}) \quad \Longrightarrow \quad T = \frac{\log 2}{k}.$$

Now two particles enter simultaneously from opposite ends, initially separated by $L > 0$, with inward speeds $u_1, u_2 > 0$. Their distances into the block at time t are

$$s_1(t) = \frac{u_1}{k}(1 - e^{-kt}), \qquad s_2(t) = \frac{u_2}{k}(1 - e^{-kt}).$$

Finite-time collision requires $s_1 + s_2 = L$. Since $1 - e^{-kt} < 1$ for every finite $t \geq 0$, the necessary and sufficient condition is

$$\boxed{u_1 + u_2 > kL}.$$

When $u_1 + u_2 = kL$, the separation tends to zero but is positive at every finite time; that is a limiting meeting, not a finite-time collision. This corrects the non-strict inequality in both supplied versions. For a finite collision,

$$T = \frac{1}{k}\log\frac{u_1 + u_2}{u_1 + u_2 - kL}, \qquad s_1(T) = \frac{u_1 L}{u_1 + u_2}.$$

The second expression gives the collision position measured from the first particle's entry end. As $u_1 + u_2$ decreases to kL from above, the collision time diverges, confirming the strict threshold.

Method. For motion: choose the positive direction first; draw forces; write the signed net force; choose dv/dt or $v\,dv/dx$ according to the relation wanted; include initial conditions and equilibrium velocities; check the result's units, signs and limiting behaviour.